National University of Computer and Emerging Sciences
Karachi Campus

Artificial Intelligence (AI2002)

Final Exam

Date: May 27th 2024

Total Time (Hrs): 3
Total Marks: 50
Total Questions: 7

Course Instructor(s)

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Roll No Section Student Signature

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Attempt all the questions.

CLO#1: To grasp the concepts of rational behavior and intelligent agents

Q1: Answer the following brief questions

1. What are the primary characteristics that define a rational agent in Artificial Intelligence?
   Answer: The primary characteristics that define a rational agent in artificial intelligence are: Goal-Oriented Behavior, Perception and Sensing, Decision-Making Ability, Action Execution, Adaptability and Learning, Performance Measurement, and Bounded Rationality
2. Why CSPs use factored state representation? Give justification.
   Answer: it effectively manages complexity and facilitates problem-solving in a structured and efficient manner. It provides modularity, scalability, incremental problem solving and etc.
3. An autonomous drone delivery system operates in a complex environment where it must deliver packages to various locations within a city. Explain whether the environment for the agent is episodic or sequential. Justify your answer.
   Answer: Episodic. Since each delivery is independent of the other, the environment should be considered episodic
4. How does local beam search differ from standard hill-climbing algorithms in terms of handling multiple states simultaneously during search?
   Answer: By managing multiple states, local beam search can explore the search space more thoroughly and is less likely to get trapped in local optima compared to the single-path approach of standard hill-climbing.
5. What are the key differences between depth-first search (DFS) and breadth-first search (BFS) in terms of space and time complexity?
   Answer: Both DFS and BFS have the same time complexity, their space complexity and traversal strategies differ significantly, making each suitable for different types of problems and graph structures.
6. Assume that a rook can move on a chessboard any number of squares in a straight line, vertically or horizontally, but cannot jump over other pieces. Manhattan distance is an admissible heuristic for the problem of moving the rook from square A to square B in the smallest number of moves. Justify.
   Answer: Since the Manhattan distance reflects the minimum travel required horizontally and vertically, and since the rook can cover any number of squares in these directions in a single move, the Manhattan distance never overestimates the actual cost. Therefore, it is an admissible heuristic for this problem.

7. Minimax algorithm leads to outcomes at least as good as any other strategy only when the opponent is also optimal. Give Justification.
   Answer: The Minimax algorithm leads to outcomes at least as good as any other strategy when the opponent is optimal because it is designed to handle the worst-case scenario with perfect rationality. While it guarantees the best outcome under optimal play, it may not always exploit suboptimal play as effectively as other adaptive strategies might.
8. Write any two limitations of hill climbing algorithm in optimization problems
   Answer: Local Optima and Plateaus and Ridges
9. Given a CSP with n variables of domain size d, what would be the size of the search tree: (i) with commutativity (ii) Without commutativity.
   Answer: With Commutativity: Exploiting commutativity fully theoretically reduces the number of unique nodes in the search tree, but this is not typically reflected in practical CSP algorithms. The combinatorial reduction is represented by $\binom{n+d-1}{d-1}$ though it's not a standard approach in typical search algorithms.
   Without Commutativity: The search tree size is, d^n representing all possible ordered assignments.
10. Iterative deepening search can also be applied to adversarial search. Give Justification.
    Answer: Iterative deepening search is well-suited for adversarial search because it combines the memory efficiency of depth-first search with the completeness of breadth-first search, handles time constraints effectively, benefits from improved move ordering, and integrates seamlessly with alpha-beta pruning to enhance search efficiency. **[10 Marks]**
    Rubrics: 0.5 – Partial Attempt, 1 – Accurate Answer

CLO#3: To showcase comprehension and capability in implementing key concepts and methodologies in probabilistic reasoning, machine learning algorithms, and neural networks

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Q2: (A). Consider a population in which 30% of individuals have a certain genetic trait, denoted as Trait A. Among those with Trait A, 70% also have Trait B, while among those without Trait A, 20% have Trait B.

1. What is the conditional probability that an individual has Trait B given that they have Trait A?
   Solution: $\boldsymbol{P(B|A) = 70\% = 0.7}$
2. What is the unconditional probability that a randomly selected individual from the population has Trait B?
   Solution:
   Given: $\boldsymbol{P(A) = 30\% = 0.3}$
   $\boldsymbol{P(-A) = 1 - P(A) = 1 - 0.3 = 0.7}$
   $\boldsymbol{P(B|A) = 0.7}$
   $\boldsymbol{P(B|-A) = 0.2}$
   To find P(B), we use the law of total probability, which states
   $\boldsymbol{P(B) = P(B|A)P(A) + P(B|-A)P(-A)}$
   $\boldsymbol{P(B) = (0.7 * 0.3) + (0.2 * 0.7)}$
   $\boldsymbol{P(B) = (0.21) + (0.14)}$
   $\boldsymbol{P(B) = 0.35}$

(B). Calculate the Joint Probability Distribution of the following figure 1.

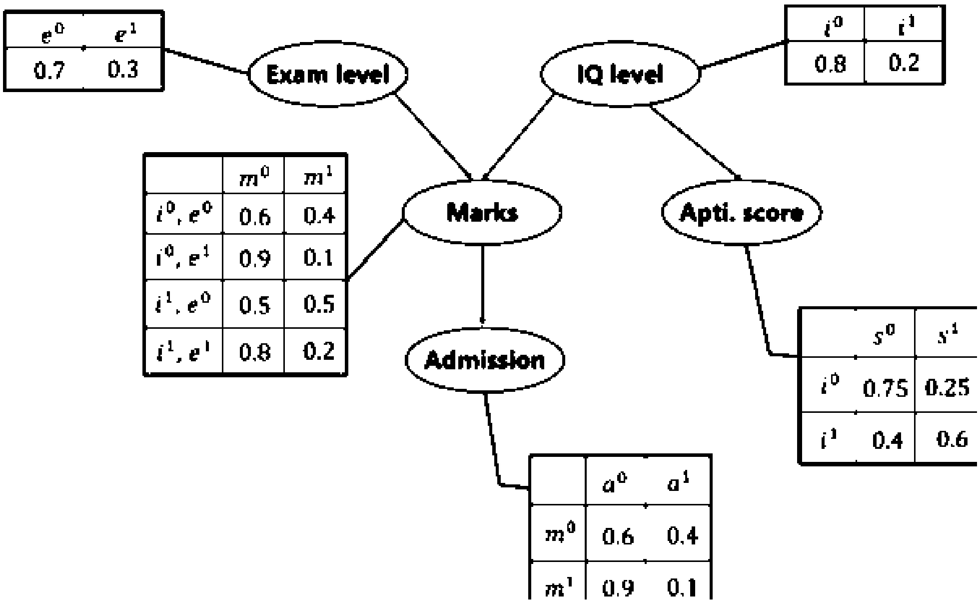

Figure 1: Bayesian Network

Solution:

Given Variables:

Exam Level (e), IQ level (i), Aptitude Score (s), Marks (m), Admission (a)

Formula: $P[a, m, i, e, s] = P(a|m).P(m|i, e).P(i).P(e).P(s|i)$

1. Calculate the probability that in spite of the exam level being difficult, the student having a low IQ level and a low Aptitude Score, manages to pass the exam and secure admission to the university.
   Solution: $P[a = 1, m = 1, i = 0, e = 1, s = 0] =$
   $P(a = 1|m = 1).P(m = 1|i = 0, e = 1).P(i = 0).P(e = 1).P(s = 0|i = 0)$
   $= 0.1 * 0.1 * 0.8 * 0.3 * 0.75 = 0.0018$

2. In another case, calculate the probability that the student has a High IQ level and Aptitude Score, the exam being easy yet fails to pass and does not secure admission to the university.
   Solution: $P[a = 0, m = 0, i = 1, e = 0, s = 1] =$
   $P(a = 0|m = 0).P(m = 0|i = 1, e = 0).P(i = 1).P(e = 0).P(s = 1|i = 1)$
   $= 0.6 * 0.5 * 0.2 * 0.7 * 0.6 = 0.0252$

[2+2+2 = 6 Marks]

Q3: (A). How does Naïve Bayes handle feature independence and missing data?
Solution: Naïve Bayes assumes that the features are conditionally independent given the class label. This means that the presence or absence of a particular feature is assumed to be unrelated to the presence or absence of any other feature, given the class label. One common approach for handling missing data in Naïve Bayes is to ignore the missing features during the calculation of the conditional probabilities.

(B). Build a Naïve Bayes Classifier for the given table 1 and use your model to classify the following table 2 as new instances.

Instance	Education Level	Career	Years of Experience	Salary
1	High School	Management	Less than 3	Low
2	High School	Management	3 to 10	Low
3	College	Management	Less than 3	High
4	College	Service	More than 10	Low
5	High School	Service	3 to 10	Low
6	College	Service	3 to 10	High
7	College	Management	More than 10	High
8	College	Service	Less than 3	Low
9	High School	Management	More than 10	High
10	High School	Service	More than 10	Low

Table 1: Training Data

Instance	Education Level	Career	Years of Experience
1	High School	Service	Less than 3
2	College	Retail	Less than 3
3	Graduate	Service	3 to 10

Table 2: Test Data

Solution:

Step1: Calculate Prior Probabilities

From Table 1: Total Instances: 10, Salary = Low:5, Salary = High:5

Prior Probabilities: $\boldsymbol{P(Low) = \frac{5}{10} = 0.5, P(High) = \frac{5}{10} = 0.5}$

Step 2: Calculate Likelihoods

For Salary = Low

Education Level: High School: 3/5 = 0.6, College: 2/5 = 0.4, Graduate = 0/5 = 0 (not present)

Career: Management: 2/5 = 0.4, Service: 3/5 = 0.6, Retail: 0/5 = (not present)

Years of Experience: Less than 3: 2/5 = 0.4, 3 to 10: 2/5 = 0.4, More than 10: 1/5 = 0.2

For Salary = High

Education Level: High School: 1/5 = 0.2, College: 4/5 = 0.8, Graduate: 0/5 = 0 (not present)

Career: Management: 3/5 = 0.6, Service: 2/5 = 0.4, Retail: 0/5 = 0 (not present)

Years of Experience: Less than 3: 1/5 = 0.2, 3 to 10: 2/5 = 0.4, More than 10: 2/5 = 0.4

Step 3: Classify New Instances

Instance 1: High School, Service, Less than 3

For Low: $\boldsymbol{P(Low|Instance\ 1)} \propto$
$\boldsymbol{P(Low)\ .\ P(High\ School|Low).\ P(Service|Low).\ P(Less\ than\ 3|Low)}$
$\propto \boldsymbol{0.5\ x\ 0.6\ x\ 0.6\ x\ 0.4 = 0.072}$

For High: $\boldsymbol{P(High|Instance\ 1)} \propto$
$\boldsymbol{P(High)\ .\ P(High\ School|High).\ P(Service|High).\ P(Less\ than\ 3|High)}$
$\propto \boldsymbol{0.5\ x\ 0.2\ x\ 0.4\ x\ 0.2 = 0.008}$

Since 0.072 > 0.008 , classify as low

Instance 2: College, Retail, Less than 3

For Low: $\boldsymbol{P(Low|Instance\ 2)} \propto$
$\boldsymbol{P(Low)\ .\ P(College|Low).\ P(Retail|Low).\ P(Less\ than\ 3|Low)}$
$\propto \boldsymbol{0.5\ x\ 0.4\ x\ 0\ x\ 0.4 = 0}$

For High: $\boldsymbol{P(High|Instance\ 2)} \propto$
$\boldsymbol{P(High)\ .\ P(College|High).\ P(Retail|High).\ P(Less\ than\ 3|High)}$
$\propto \boldsymbol{0.5\ x\ 0.8\ x\ 0\ x\ 0.2 = 0}$

Both are 0 due to retail having 0 probability in both classes. Hence we'll consider it undefined

Instance 3: Graduate, Service, 3 to 10

For Low: $P(Low|Instance\ 3) \propto$
$P(Low)\ .P(Graduate|Low).P(Service|Low).P(3\ to\ 10|Low)$
$\propto \mathbf{0.5\ x\ 0\ x\ 0.6\ x\ 0.4 = 0}$

For High: $P(High|Instance\ 3) \propto$
$P(High)\ .P(Graduate|High).P(Service|High).P(3\ to\ 10|High)$
$\propto \mathbf{0.5\ x\ 0\ x\ 0.4\ x\ 0.4 = 0}$

Both are 0 due to Graduate having 0 probability in both classes. Hence we'll consider it undefined.

[2 + 4 = 6 Marks]

Q4: (A). List down two applications of regression and correlation.

Answer:

Regression: Predictive Analysis in Finance, Health Care
Correlation: Market Research, Environmental Studies

(B). The yield of a chemical process is a random variable whose value is considered to be a linear function of the temperature x. The following data in table 3 of corresponding values of x and y.

Temperature in C° (x)	0	25	50	75	100	125
Yield in Grams (y)	10	25	40	55	70	?

Table 3: Training Data

Find the estimated yield in grams using regression, if the temperature is raised to 125 C°

Solution: Solution is attached in file named as Q4 Part B Solution.

[1 + 3 = 4 Marks]

Q5: (A). You are a data analyst at a healthcare organization. Your task is to predict whether a patient is likely to have diabetes based on historical patient data. The dataset includes the following features for each patient:
- Age - BMI (Body Mass Index) - Blood Pressure - Insulin Level - Glucose Level
- Diabetes status (1 for diabetes, 0 for no diabetes)

Given the following data in table 4, use the K-Nearest Neighbors (KNN) algorithm to classify whether a new patient with the following characteristics will have diabetes, use k=3:

Age:	BMI:	Blood Pressure	Insulin Level	Glucose Level
45	28	80	100	140

Table 4: Test Data

The historical data is in table 5 as follows:

Age	BMI	Blood Pressure	Insulin Level	Glucose Level	Diabetes
25	22	70	85	130	0
30	30	85	90	150	1
35	28	80	105	160	1
40	25	75	100	140	0
50	27	78	95	145	0
55	31	90	110	155	1
60	26	85	108	148	0
65	29	88	115	165	1
70	32	92	120	170	1

Table 5: Training Data

Tasks:
1. Calculate the Euclidean distance between the new patient and all patients in the dataset.
2. Identify the K-nearest neighbors based on the distances calculated.
3. Predict whether the new patient will have diabetes using majority voting.
4. Discuss how the choice of k affects the classification result and how you might select an optimal k.

Solution:
Part 1: Identify the Features and the Target Variable
The features are:
- Age
- BMI
- Blood Pressure
- Insulin Level
- Glucose Level
The target variable is:
- Diabetes status

Part 2: Calculate the Euclidean Distance
Using the Euclidean distance formula:

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2 + (w_2 - w_1)^2 + (v_2 - v_1)^2}$$

Calculate the distances between the new patient and each patient in the dataset:

- To (25, 22, 70, 85, 130):

  $$\sqrt{(45-25)^2 + (28-22)^2 + (80-70)^2 + (100-85)^2 + (140-130)^2} = \sqrt{20^2 + 6^2$$

- To (30, 30, 85, 90, 150):

  $$\sqrt{(45-30)^2 + (28-30)^2 + (80-85)^2 + (100-90)^2 + (140-150)^2} = \sqrt{15^2 + (-$$

- Similarly, calculate for other data points.

Part 3: Identify the K-Nearest Neighbors

Based on the calculated distances, identify the three nearest neighbors. The calculated distances are:

Patient	Distance
1	29.34
2	21.31
3	15.62
4	5.00
5	10.95
6	20.00
7	24.08
8	32.60
9	36.25

The three smallest distances are from patients 4, 5, and 2.

Part 4: Predict Using Majority Voting
Check the 'Diabetes' status of the 3 nearest neighbors:
- Patient 4: 0 (no diabetes)
- Patient 5: 0 (no diabetes)
- Patient 2: 1 (diabetes)

Since 2 out of 3 nearest neighbors do not have diabetes, the new patient is predicted to not have diabetes.
Final Prediction:
Thus, based on the K-Nearest Neighbors algorithm, the new patient is predicted not to have diabetes.

Part 5: Discussion on the choice of k and selecting an optimal k

- Impact of k on classification:
    - A smaller k (e.g., k=1) makes the classifier sensitive to noise, as it only considers the closest neighbor.
    - A larger k (e.g., k=10) smooths out noise but may include too many points from other classes, leading to potential misclassification.
    - An odd k is often chosen to avoid ties in binary classification problems.
- Selecting an optimal k:
    - Cross-validation: Split the dataset into training and validation sets multiple times and choose the k that minimizes the error rate on the validation sets.
    - Heuristics: Some heuristics suggest using √n where nnn is the number of data points.
    - Domain knowledge: Understanding the specific problem domain might guide the choice of k.

[1.5 + 1 + 1 + 0.5 = 4 Marks]

Q5: (B). Construct a complete decision tree using the ID3 algorithm on the provided data. Display the entire tree from the root node to leaf nodes as indicated in the set. Dataset of animal classification have the following attributes:

- Habitat (Forest, Desert, Grassland)
- Diet (Herbivore, Carnivore, Omnivore)
- Activity (Diurnal, Nocturnal)

Animal Type (Mammal, Bird, Reptile) - this is the target variable. **[8 Marks]**

Habitat	Diet	Activity	Animal Type
Forest	Herbivore	Diurnal	Mammal
Forest	Carnivore	Nocturnal	Mammal
Desert	Carnivore	Nocturnal	Reptile
Grassland	Herbivore	Diurnal	Bird
Forest	Omnivore	Nocturnal	Mammal
Grassland	Omnivore	Diurnal	Bird
Desert	Omnivore	Nocturnal	Reptile
Grassland	Herbivore	Diurnal	Bird
Forest	Herbivore	Nocturnal	Mammal
Desert	Herbivore	Diurnal	Reptile

Table 6: Training Data

Solution: The difference I made in this question that I shrink the data up-till 10 rows, so student's solution has minor diff of calculation that they place 10 in place of 14.

1. **Calculate Entropy of the Target Variable**

First, calculate the entropy of the target variable "Animal Type".

$$\text{Entropy}(S) = -\sum_{i=1}^{c} p_i \log_2(p_i)$$

Where p_i is the proportion of each class in the target variable.

Counts of Animal Type:

- Mammal: 5
- Bird: 5
- Reptile: 4

$$\text{Entropy}(S) = -\left(\frac{5}{14}\log_2\frac{5}{14} + \frac{5}{14}\log_2\frac{5}{14} + \frac{4}{14}\log_2\frac{4}{14}\right)$$
$$= -(0.357\log_2 0.357 + 0.357\log_2 0.357 + 0.286\log_2 0.286)$$
$$= -(0.357 \times -1.485 + 0.357 \times -1.485 + 0.286 \times -1.807)$$
$$= 1.556$$

2. **Calculate Information Gain for Each Attribute**

To decide which attribute to split on, we calculate the Information Gain for each attribute.

Attribute: Habitat

$$\mathrm{IG}(S, \mathrm{Habitat}) = \mathrm{Entropy}(S) - \sum_{v \in \mathrm{Values}(\mathrm{Habitat})} \frac{|S_v|}{|S|} \mathrm{Entropy}(S_v)$$

Values of Habitat:

- Forest
- Desert
- Grassland

Subset Entropies:

- **Forest:**
  - Counts: Mammal: 4, Bird: 0, Reptile: 0
  - Entropy(Forest) = 0 (since all are Mammals)
  - Probability: $\frac{5}{14}$

- **Desert:**
  - Counts: Mammal: 0, Bird: 0, Reptile: 4
  - Entropy(Desert) = 0 (since all are Reptiles)
  - Probability: $\frac{4}{14}$
- **Grassland:**
  - Counts: Mammal: 1, Bird: 5, Reptile: 0
  - Entropy(Grassland) = -\left(\frac{5}{6} \log_2 \frac{5}{6} + \frac{1}{6} \log_2 \frac{1}{6} \right)
  - Entropy(Grassland) = 0.65
  - Probability: $\frac{5}{14}$

$$\mathrm{IG}(S, \mathrm{Habitat}) = 1.556 - \left(\frac{5}{14} \times 0 + \frac{4}{14} \times 0 + \frac{5}{14} \times 0.65 \right)$$

$$= 1.556 - (0 + 0 + 0.232) = 1.324$$

Attribute: Diet

Values of Diet:

- Herbivore
- Carnivore
- Omnivore

Subset Entropies:

- **Herbivore:**
  - Counts: Mammal: 3, Bird: 0, Reptile: 1
  - Entropy(Herbivore) = 0.811
  - Probability: $\frac{4}{14}$
- **Carnivore:**
  - Counts: Mammal: 2, Bird: 1, Reptile: 3
  - Entropy(Carnivore) = 1.251
  - Probability: $\frac{5}{14}$
- **Omnivore:**
  - Counts: Mammal: 2, Bird: 2, Reptile: 1
  - Entropy(Omnivore) = 1.370
  - Probability: $\frac{5}{14}$

$$\mathrm{IG}(S, \mathrm{Diet}) = 1.556 - \left(\frac{4}{14} \times 0.811 + \frac{5}{14} \times 1.251 + \frac{5}{14} \times 1.370\right)$$
$$= 1.556 - (0.232 + 0.446 + 0.489) = 0.389$$

Attribute: Activity

Values of Activity:

- Diurnal
- Nocturnal

Subset Entropies:

- **Diurnal:**
  - Counts: Mammal: 2, Bird: 3, Reptile: 2
  - Entropy(Diurnal) = 1.371
  - Probability: $\frac{7}{14}$

- **Nocturnal:**
  - Counts: Mammal: 3, Bird: 2, Reptile: 2
  - Entropy(Nocturnal) = 1.329
  - Probability: $\frac{7}{14}$

$$\mathrm{IG}(S, \mathrm{Activity}) = 1.556 - \left(\frac{7}{14} \times 1.371 + \frac{7}{14} \times 1.329\right)$$

$$= 1.556 - (0.686 + 0.664) = 0.206$$

From the information gain calculations, **Habitat** provides the highest information gain.

3. **Build the Decision Tree**

Root Node: Habitat

- **Forest** (all Mammals): Leaf Node - Mammal
- **Desert** (all Reptiles): Leaf Node - Reptile
- **Grassland**: Split further

3. **Build the Decision Tree**

Root Node: Habitat

- **Forest** (all Mammals): Leaf Node - Mammal
- **Desert** (all Reptiles): Leaf Node - Reptile
- **Grassland**: Split further

For Grassland:

Calculate IG for remaining attributes (Diet, Activity):

- **Diet**
  - Herbivore: Bird
  - Carnivore: Bird
  - Omnivore: Bird

Since all are Birds, we can terminate the tree for Grassland as a leaf node.

```
      Habitat
     /   |    \
  Forest Desert Grassland
    |      |        |
 Mammal Reptile    Bird
```

Q6: Use the k-means algorithm and Euclidean distance to cluster the following 8 examples into 3 clusters:

Data Points	A1	A2	A3	A4	A5	A6	A7	A8
x	2	2	8	5	7	6	1	4

Suppose that the initial seeds (centers of each cluster) are A1, A4 and A7. Initially, run the k-means algorithm for 2 epochs. After the 2nd epoch, please give the following:

a) New Centroid Values
b) Newly formed cluster sets.
c) How many more iterations are needed to converge? Please Justify. **[2+2+2= 6 Marks]**

Solution:

Solution:
a)
d(a,b) denotes the Eucledian distance between a and b. It is obtained directly from the distance matrix or calculated as follows: d(a,b)=sqrt($(x_b-x_a)^2+(y_b-y_a)^2$))
seed1=A1=(2,10), seed2=A4=(5,8), seed3=A7=(1,2)

epoch1 – start:

A1:
d(A1, seed1)=0 as A1 is seed1
d(A1, seed2)= $\sqrt{13}$ >0
d(A1, seed3)= $\sqrt{65}$ >0
➔A1 $\in$ cluster1

A2:
d(A2,seed1)= $\sqrt{25}$ = 5
d(A2, seed2)= $\sqrt{18}$ = 4.24
d(A2, seed3)= $\sqrt{10}$ = 3.16 ← smaller
➔ A2 $\in$ cluster3

A3:
d(A3, seed1)= $\sqrt{36}$ = 6
d(A3, seed2)= $\sqrt{25}$ = 5 ← smaller
d(A3, seed3)= $\sqrt{53}$ = 7.28
➔ A3 $\in$ cluster2

A4:
d(A4, seed1)= $\sqrt{13}$
d(A4, seed2)=0 as A4 is seed2
d(A4, seed3)= $\sqrt{52}$ >0
➔ A4 $\in$ cluster2

A5:
d(A5, seed1)= $\sqrt{50}$ = 7.07

A6:
d(A6, seed1)= $\sqrt{52}$ = 7.21

d(A5, seed2)= $\sqrt{13}$ = 3.60 ← smaller
d(A5, seed3)= $\sqrt{45}$ = 6.70
→ A5 $\in$ cluster2

d(A6, seed2)= $\sqrt{17}$ = 4.12 ← smaller
d(A6, seed3)= $\sqrt{29}$ = 5.38
→ A6 $\in$ cluster2

A7:
d(A7, seed1)= $\sqrt{65}$ >0
d(A7, seed2)= $\sqrt{52}$ >0
d(A7, seed3)=0 as A7 is seed3
→ A7 $\in$ cluster3

A8:
d(A8, seed1)= $\sqrt{5}$
d(A8, seed2)= $\sqrt{2}$ ← smaller
d(A8, seed3)= $\sqrt{58}$
→ A8 $\in$ cluster2

end of epoch1

new clusters: 1: {A1}, 2: {A3, A4, A5, A6, A8}, 3: {A2, A7}

b) centers of the new clusters:
C1= (2, 10), C2= ((8+5+7+6+4)/5, (4+8+5+4+9)/5) = (6, 6), C3= ((2+1)/2, (5+2)/2) = (1.5, 3.5)

Note: 2nd iteration details are yet to be shown here, however the new clusters and their centers are mentioned in the details below.

At the end of 2nd epoch show: a) The new clusters (i.e. the examples belonging to each cluster)

After the 2nd epoch the results would be:

1: {A1, A8},

2: {A3, A4, A5, A6},

3: {A2, A7}

b) The centers of the new clusters After the 2nd epoch the new centers would be: centers C1=(3, 9.5), C2=(6.5, 5.25) and C3=(1.5, 3.5).

c) How many more iterations are needed to converge? We would need one more epoch for the points to converge. After the 3rd epoch, the results would be:

1: {A1, A4, A8},

2: {A3, A5, A6},

3: {A2, A7}

with centers C1=(3.66, 9), C2=(7, 4.33) and C3=(1.5, 3.5)

Q7: Consider a feedforward neural network with one input layer consisting of two input neurons, two hidden layers with three neurons each, and one output layer with one neuron. The activation function used in all layers is the sigmoid function. Draw the neural network and compute the output.

Input Layer to First Hidden Layer (Layer 1):	First Hidden Layer to Second Hidden Layer (Layer 2):	Second Hidden Layer to Output Layer (Layer 3):
Neuron 1: Weight 1: 0.1, Bias: -0.2 Weight 2: -0.3, Bias: 0.4 Neuron 2: Weight 1: 0.5, Bias: -0.1 Weight 2: -0.2, Bias: 0.3 Neuron 3: Weight 1: 0.4, Bias: 0.2 Weight 2: -0.1, Bias: -0.5	Neuron 1: Weight 1: 0.2, Bias: 0.1 Weight 2: -0.4, Bias: 0.3 Weight 3: 0.5, Bias: -0.2 Neuron 2: Weight 1: 0.3, Bias: -0.4 Weight 2: -0.2, Bias: 0.2 Weight 3: 0.1, Bias: 0.3 Neuron 3: Weight 1: -0.1, Bias: 0.2 Weight 2: 0.4, Bias: -0.3 Weight 3: -0.2, Bias: 0.1	Neuron 1: Weight 1: -0.3, Bias: 0.5 Weight 2: 0.2, Bias: -0.1 Weight 3: 0.4, Bias: 0.3

Given an input vector [0.6, -0.1], calculate the output value of the neural network. Assume the sigmoid activation function for all neurons. **[6 Marks]**

Solution:

1. Input Layer to First Hidden Layer (Layer 1):

   For each neuron in the first hidden layer, we calculate the weighted sum of inputs and add the bias, then apply the sigmoid activation function.

   Neuron 1: $z1(1) = (0.6 * 0.1) + (-0.1 * -0.3) + (-0.2) = 0.06 + 0.03 - 0.2 = -0.11 z1(1) = (0.6 * 0.1) + (-0.1 * -0.3) + (-0.2) = 0.06 + 0.03 - 0.2 = -0.11$ $a1(1) = sigmoid(-0.11) \approx 0.472 a1(1) = sigmoid(-0.11) \approx 0.472$
   Neuron 2: $z2(1) = (0.6 * 0.5) + (-0.1 * -0.2) + (-0.1) = 0.3 + 0.02 - 0.1 = 0.22 z2(1) = (0.6 * 0.5) + (-0.1 * -0.2) + (-0.1) = 0.3 + 0.02 - 0.1 = 0.22$ $a2(1) = sigmoid(0.22) \approx 0.554 a2(1) = sigmoid(0.22) \approx 0.554$
   Neuron 3: $z3(1) = (0.6 * 0.4) + (-0.1 * -0.1) + (0.2) = 0.24 + 0.01 + 0.2 = 0.45 z3(1) = (0.6 * 0.4) + (-0.1 * -0.1) + (0.2) = 0.24 + 0.01 + 0.2 = 0.45$ $a3(1) = sigmoid(0.45) \approx 0.611 a3(1) = sigmoid(0.45) \approx 0.611$

2. First Hidden Layer to Second Hidden Layer (Layer 2):

Similar to the previous step, we calculate the weighted sum of inputs plus the bias for each neuron in the second hidden layer and apply the sigmoid activation function.
Neuron 1: $z1(2) = (0.472 * 0.2) + (0.554 * -0.4) + (0.611 * 0.5) + 0.1 = 0.0944 - 0.2216 + 0.3055 + 0.1 = 0.2783 z1(2) = (0.472 * 0.2) + (0.554 * -0.4) + (0.611 * 0.5) + 0.1 = 0.0944 - 0.2216 + 0.3055 + 0.1 = 0.2783$ $a1(2) = sigmoid(0.2783) \approx 0.569 a1(2) = sigmoid(0.2783) \approx 0.569$

Neuron 2: $z2(2) = (0.472 * 0.3) + (0.554 * -0.2) + (0.611 * 0.1) - 0.4 = 0.1416 - 0.1108 + 0.0611 - 0.4 = -0.3071 z2(2) = (0.472 * 0.3) + (0.554 * -0.2) + (0.611 * 0.1) - 0.4 = 0.1416 - 0.1108 + 0.0611 - 0.4 = -0.3071$ $a2(2) = sigmoid(-0.3071) \approx 0.423 a2(2) = sigmoid(-0.3071) \approx 0.423$

Neuron 3: $z3(2) = (0.472 * -0.1) + (0.554 * 0.4) + (0.611 * -0.2) + 0.2 = -0.0472 + 0.2216 - 0.1222 + 0.2 = 0.2522 z3(2) = (0.472 * -0.1) + (0.554 * 0.4) + (0.611 * -0.2) + 0.2 = -0.0472 + 0.2216 - 0.1222 + 0.2 = 0.2522$ $a3(2) = sigmoid(0.2522) \approx 0.562 a3(2) = sigmoid(0.2522) \approx 0.562$

3. Second Hidden Layer to Output Layer (Layer 3):

Finally, for the output neuron, we calculate the weighted sum of inputs plus the bias and apply the sigmoid activation function.

$$z1(3) = (0.569 * -0.3) + (0.423 * 0.2) + (0.562 * 0.4) + 0.5 = -0.1707 + 0.0846 + 0.2248 + 0.5 = 0.6387 z1(3) = (0.569 * -0.3) + (0.423 * 0.2) + (0.562 * 0.4) + 0.5 = -0.1707 + 0.0846 + 0.2248 + 0.5 = 0.6387 \; a1(3) = sigmoid(0.6387) \approx 0.654 a1(3) = sigmoid(0.6387) \approx 0.654$$

So, the output value of the neural network for the given input vector [0.6,–0.1][0.6,–0.1] is approximately 0.654